WARNING
Ecotoxic

STINGER 450
HERBICIDE

MINISTRY FOR PRIMARY INDUSTRIES
NEW ZEALAND
30/07/2024

A non-selective herbicide for the control of many annual and perennial weeds in a wide range of situations

ACTIVE INGREDIENT:
450 g/litre glyphosate as the isopropylamine salt in the form of a soluble concentrate.

For the control of gorse, broom, blackberry and other brushweeds and certain broadleaf weeds in pasture, forestry, amenity turf and non-cropland areas.

READ THE LABEL COMPLETELY BEFORE MIXING OR APPLYING.

NET CONTENTS: 5L – 1000L

EMERGENCY RESPONSE (ALL HOURS)
0800 CHEMCALL or 0800 243 622.
Contact PGG Wrightson Ltd Freephone 0800 10 22 76

AGRECOVERY

Registered to:
PGG Wrightson Ltd
1 Robin Mann Place
Christchurch Airport 8053
PO Box 292
Christchurch 8140
www.pggwrightson.co.nz

Version3 - May 2024

READ THE LABEL COMPLETELY BEFORE USE

HAZARD CLASSIFICATION: Hazardous to the aquatic environment (chronic) Category 2

ECOTOXICITY

WARNING: Toxic to some aquatic plants.
Do not apply directly into or onto water.
To be mixed, loaded and applied by trained and qualified person.

HANDLING PRECAUTIONS

STORAGE

Keep out of reach of children. Store in the original, tightly closed container, in a secure cool, dry place out of direct sunlight and away from seed, fertilizers, feed or foodstuffs.
Storage of certain quantities of this product trigger specific storage requirements.

PERSONAL PROTECTION

Avoid contact with eyes and skin. Avoid inhalation of spray mist. When handling, mixing or applying, wear overalls, boots, gloves and eye protection. Clean protective equipment daily after work. Do not eat, drink or smoke while using. Remove protective clothing and wash hands and face thoroughly before meals and after work.

EQUIPMENT

Apply with well-maintained and calibrated spray equipment in accordance with The New Zealand Standard for the Management of Agrichemicals (NZS 8409:2004).

DISPOSAL

CONTAINER

Triple rinse empty container and add residue to spray tank. Recycle empty container through Agrecovery (0800 247 326, www.agrecovery.co.nz). Otherwise, crush and bury in a suitable landfill. DO NOT reuse this container for any other purpose.

PRODUCT

Dispose of this product only by using according to the label, or at an approved landfill or other approved facility.

ENVIRONMENT

Use this product carefully. Do not contaminate aquatic environments with product, spray drift or empty container. Spray drift or equipment contamination may cause serious damage to desirable plants. Do not allow spray drift outside the target area to occur. If accidental spraying occurs, immediately wash spray off foliage with plenty of water to minimise damage.

Spillage

Exclude non-essential people from the area. Wear protective clothing and prevent material from entering waterways. Absorb spill with inert material such as soil or sand, and place in waste container. Wash area with water, and absorb with further inert material. Dispose of safely to a suitable landfill.

First Aid

If swallowed do not induce vomiting. If splashed in eyes, wash out immediately with water. Get medical advice. If skin contact occurs, remove contaminated clothing and wash with running water. For advice, call the National Poisons Centre 0800 POISON (0800 764 766) (24 hrs) or a doctor.

GENERAL INFORMATION

STINGER 450 is a non-selective glyphosate herbicide for use in agriculture and horticulture.

- Before planting crops and trees
- In pasture renovation and reduced tillage systems
- For pre-harvest weed control
- In vineyards, orchards and market gardens
- On turf and recreational areas
- In drains and waterways.

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STINGER 450 does not have adverse ecotoxic effects on fish or crustacea.
STINGER 450 is water soluble and is absorbed by green plant tissue. It is then translocated throughout the plant, including the root system, to kill the entire plant. Plants should be actively growing, and not under drought stress, with clean foliage at the time of spraying.
The product degrades in the soil and has no residual weed control activity.
Rainfall within 2 hours of application may reduce effectiveness. Application to foliage which is already wet will increase the time required for rainfastness.
Visible effects may not be apparent for 14 days depending on weather conditions. Some perennial weeds may not show effects for 2 - 3 weeks.

DIRECTIONS FOR USE

HSNO Controls:

The maximum application rate of this substance is 14.4 L/ha (equivalent to 7.2 kg of glyphosate /ha) for aerial and ground-based application methods on land.
The maximum application frequency of this substance on land must not be more than two per calendar year with minimum interval period of 3 months.
The maximum application rate of this substance is 4.5 L/ha (equivalent to 2.25 kg of glyphosate /ha) for aerial and ground-based application methods into or onto water.
The maximum application frequency of this substance into or onto water must not be more than once per calendar year with minimum interval period of one year.

Application Methods

STINGER 450 must only be placed in equipment made from stainless steel, fibreglass, plastic, aluminium, brass or copper. Other types of container may cause a reaction in which flammable hydrogen gas is generated. All spray equipment including pumps, spray tanks, lines and nozzles, and landing gear of aircraft should be thoroughly washed with water after each spraying day.
Boom equipment: Application can be made in spray volumes up to 200 litres/ha using fan nozzles at pressures in the range of 200-280kPa. Boom height should be set so that a double overlap of nozzle pattern is achieved at the top of the target plants.

Aerial equipment: Applicable to forest and non-crop areas only. Can be applied by aircraft in spray volumes of up to 100 litres/ha. Increase spray volumes in hilly terrain to ensure maximum target contact. Do not use in intensive horticultural cropping areas.
Handgun and Knapsack: Ensure a fine spray pattern to give complete coverage. For handguns, use pressures of 1000 - 2000 kPa and No. 4 and No. 6 spray tips
Mistblower: Refer to table.

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Mixing

Fill the clean spray tank with half the required amount of clean water. Add STINGER 450 as required and mix well before adding the remaining water. If surfactant is required, add this last to minimise foaming. Agitate well before spraying. When tank-mixing with other chemicals, add STINGER 450 after other water- soluble concentrates, but before suspension concentrates. Soil particles and hard water containing calcium salts deactivate STINGER 450. If hard water cannot be avoided, the addition of a surfactant such as a pH buffer is recommended.

Compatibility

STINGER 450 is compatible with commonly used herbicides.

WITHHOLDING PERIODS

It is an offence for users of this product to cause residues exceeding the relevant MRL in the Food Notice: Maximum Residue Levels for Agricultural Compounds.

Wheat, barley, oat grain - 7 days

Animal Safety: Poisonous plants, such as ragwort, may become more palatable to stock after spraying, and livestock should be kept away until plants have died down. Otherwise, withholding is not required for stock grazing. However, it is recommended that grazing is delayed for 3 days for annual weeds, and 7 days for perennial weeds. This will allow STINGER 450 to penetrate the weeds for maximum effectiveness.

WEED CONTROL NOTES

Agricultural Areas: May be used for the control of annual and perennial weeds as indicated in the Weeds Controlled Table, prior to sowing any crop.
Minimum Tillage: For total weed control prior to seeding in a reduced tillage system. Weeds should be actively growing and not under stress from frost, drought or waterlogging. In pastures where only annual weeds are present use 800mL/ha to control seedling annual weeds and 2.4 litres/ha to control mature annual weeds. For perennial weeds the higher rate of 2.4 litres/ha with the addition of organo-silicone surfactant at 200mL/100 litres is recommended. For the improved control of white clover, tank mix with a dicamba 200 g/L product at 1.5 litres/ha. Do not plant legume crops within 28 days of applying dicamba. Cultivation may start from 7 days after spraying perennial and 3 days for annual weeds. Cultivation should occur within 21 days after spraying. Do not sow until a satisfactory seedbed is formed.
Pasture: Spot application to specific weeds. May be applied to control a wide range of weeds in pasture. It is non-selective and may damage or kill any plant in the spray area.
Pre-Harvest Weed Control: Wheat, barley, oats. Spray 7-12 days prior to harvest when the grain moisture is less than 30%. Do not harvest within 7 days after spraying. Do not spray undersown crops. Do not spray crops that will be harvested for seed.
Do not use as a pre-harvest desiccant.
Do not apply more than 3.2 kg/ha.

Orchards/Vines: Crops – citrus, apple, pear, stonefruit, kiwifruit, grapes. Apply as a directed or shielded spray. For long term control of annual weeds STINGER 450 may be tank mixed with an approved herbicide for residual control. Do not apply to unprotected trees less than 3 years old. Do not allow spray or spray drift to contact any part of the stonefruit tree, or green bark, suckers, buds, foliage or fruit of other trees. Spray before crops bud, sucker or bloom.
Market Gardens: Applied before or after seeding but prior to crop emergence. Spraying can also be made prior to transplanting or as a post-harvest clean up. For longer-term control of seedling weeds it may be tank mixed with an approved residual herbicide such as a simazine 500 g/L product in accordance with label recommendations.
Asparagus: Pre-crop emergence and post-harvest applications. Apply prior to planting or crop emergence. Do not treat emerging spears. Application can also be made after removal of all spears but before crop regrowth occurs. In established ferns, apply as a directed or shielded spray or with wiper equipment. Do not allow spray, spray drift or the wiper surface to contact any part of the asparagus plant
Industrial and Non-Agricultural Areas: Treat specific problem weeds as indicated in the Weeds Controlled Table. Do not allow spray drift.
Recreational Areas: Lawns, turf, parks, reserves and ornamentals. Apply before the sowing and planting of these areas. In mown areas, allow sufficient regrowth before spraying. Can also be sprayed around the base of established trees or shrubs, as a directed or shielded spray or with wiper equipment. Avoid spray drift or contact of the wiper surface with any part of the shrub or tree.
Drains and Waterways: For best results ensure that 75% of the weed to be sprayed is above water level. Avoid treating silt-laden weeds. Following spraying do not submerge the weed, as this will cause the herbicide to be washed off.
Forestry: For the control of grasses and weeds as listed in the table. Can be applied for weed control prior to burning or planting of forest or nursery or around already established trees. Do not allow spray drift or spray contact with foliage or green bark of desirable trees.

RESISTANCE MANAGEMENT

GROUP 9 HERBICIDE
STINGER 450 contains glyphosate, a Group 9 herbicide belonging to the glycine chemical family, and having the EPSP synthase mode of action. Some naturally occurring weeds may become resistant when excessive use of the same herbicide occurs. To prevent or delay the development of resistance, use in tank mixes and/or rotate use with herbicides with a different mode of action. Consult HYPERLINK “http://resistance.nzpps.org/index.php”http://resistance.nzpps.org/index.php for alternative modes of action and details of resistance management strategies (if any) for the crops listed on the label.

RESISTANT WEEDS WARNING:

To minimise the risk of resistance developing, use strictly in accordance with the label directions. Since the occurrence of resistant weeds is difficult to detect prior to use PGG Wrightson accepts no liability for losses that may result from the failure of this product to control resistant weeds.

WEEDS CONTROLLED TABLE

Weed	Boom/ Aerial litres/ha	Handgun vol/100 litres	Knapsack mL/15 litres	Mistblower mL/10 litres	When to apply
Annual weeds	0.8 - 1.6	160mL	25	50	Use the higher rate for larger weeds.
Australian sedge	5 - 7.2 (aerial)	800mL	120	240	Apply from November to March. Burn or slash from 30 days after treatment.
Barberry	-	800mL	120	240	Apply from flowering to late fruit stage (Jan-Apr). Regrowth should be 1.5m tall before spraying is carried out.
Barley grass	0.8	160mL	25	50	Apply when plants are at early seed head stage (Sep-Dec).
Blackberry	7 - 16	1.2 litres	120	240	Apply from flowering fruiting stage (Dec-Apr), before frosts occur. For hard to kill blackberry, aerial application, or no follow-up action planed, use the higher rate. Allow 6-10 weeks before slashing or burning.
Boxthorn	-	1 litre	120	-	Apply to actively growing plants (Nov-Apr). Ensure complete coverage of dense thickets. Do not use extra surfactant, or treat salt damaged bushes.
Bracken/ broom	7.2 + Slikka™	800mL + Slikka™	120 + Slikka™	240 + Slikka™	Apply after bracken fronds have unfurled (mid-Jan- June), but before fronds occur. For ground application to broom anytime, but for aerial application when broom is in full leaf.
Browntop	2.4	400mL	60	120	Apply when plants are 5-10cm high. For taller plants, graze or burn and allow regrowth of 5-10cm height before treatment.
Buddleia	7.2	800mL	120	-	Apply using handgun anytime. Aerial-treat buddleia up to 2 metres high (Feb-Apr) using spray volume of 200-300 litres/ha.
Californian thistle	3.2	800mL	120	240	Apply after flower bud development. Retreatment and/or cropping or pasture improvement may be necessary to restrict re-establishment from seed. Mowing thistles 10 days after spraying may assist control.
Cleavers	1.2	400mL	60	120	Spray actively growing plants after most reach early seedhead.

Weed	Boom/ Aerial litres/ha	Handgun vol/100 litres	Knapsack mL/15 litres	Mistblower mL/10 litres	When to apply
Couch	2.5-5	800mL	120	240	For suppression, apply 2.4 litres/ha in spring and repeat in autumn. For best results, apply 3-5 litres/ha in autumn to active growth of at least 10 cm where no previous cultivation has occurred for at least 6 months.
Docks	3-7	800mL	120	[illegible]	Spray plants in spring or summer when actively growing and before seedheads have formed. Use the lower rate for seedlings and the higher rate for mature plants. In pasture situations boom spray prior to re-seeding or establishing crops to restrict seedling re-establishment.
Floating sweet grass	5	800mL	120	240	Spray actively growing plants after most reach early seedhead. Spraying plants under drought stress may give reduced results.
Indian doab	7.2	800mL	120	-	Apply at early seedhead stage before frosts occur.
Gorse	-	800mL + Slikka™	-	-	Apply anytime ensuring complete coverage of plants not stressed by drought, grazing or previous herbicide treatment.
Kikuyu/ Mercer grass	5	800mL	120	240	Apply at early seedhead stage. On dense infestations, cut and spray the regrowth.
Nodding thistle	-	800mL	120	-	When applying by water ensure plants are actively growing (Oct-Apr) when at least 10cm of the plant can be wiped. A double pass is recommended and follow-up with spot treatments if necessary.
Old man's beard	-	0.8 - 1.6 litres	240	-	Apply Nov-Mar prior to first leaf browning. Use higher rates on stems over 1 cm in diameter. If complete coverage is not possible plants should be cut at ground level in winter and the regrowth sprayed with the higher rate when plants are at least 50cm long (March).
Pampas grass/toetoe	-	800mL	120	-	Spray spring-Autumn. Reduced control may result if plants are in flower at the time of spraying. Ensure complete coverage.

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Weed	Boom/ Aerial litres/ha	Handgun vol/100 litres	Knapsack mL/15 litres	Mistblower mL/10 litres	When to apply
Paspalum	3 - 5	800mL	120	240	Spray at low rate if plants are at early seedhead stage. Use 5 litres/ha on mature vegetative plants, before onset of frosts.
Potato (self set)	2.4	400mL	60	120	Spray actively growing plants after most reach early seedhead.
Prairie grass	1.2	160mL	25	-	Apply at early seedhead stage.
Ragwort	5	800mL	120	-	Spray when actively growing and ensure there is sufficient leaf area for herbicide uptake. When applying by wiper use on plants 10-15cm above the pasture. For best results, a double pass is recommended with a follow-up spot treatment if necessary. Caution: Ragwort becomes more palatable after treatment. Keep stock away from treated areas until plants brown-off.
Ratstail	4	800mL	120	-	Apply at a spray volume of 200 litres/ ha in autumn. In dry conditions, delay spraying until sufficient green foliage is present.
Rautahi (cutty grass)	5	800mL	120	240	Spray to actively growing plants form Jan-Mar. Burning or slashing may begin 30 days after treatment following brown-off.
Rushes	5	800mL	120	240	Burning or slashing may begin 30 days after treatment following brown-off. Wiper application to be carried out in Nov-May.
Ring fern	0.8 – 2.4	240mL	40	-	For effective control, apply 2.4L/ha. For pasture manipulation, apply 0.8-1.6L/ ha from Dec/Apr. Use higher rates for dense infestations. Burn 4-6 weeks after treatment.
Ryegrass	1.2-2.4 + Slikka™	400mL + Slikka™	60 + Slikka™	-	Spray to actively growing plants 3-10cm high. Use lower rate from (perennial) Jan-Jul when cultivation is to follow spraying, and the higher rate Aug-Dec.
Sweet brier	9.6 + Slikka™	800mL + Slikka™	120 + Slikka™	-	Spray when plant is in late flowering to early leaf yellowing stage. Ensure full coverage.
Tall fescue	5	800mL	120	240	Apply Jan-Mar to active growth.
Willow	7	800mL	-	-	Apply before leaf fall begins (Jan-Apr). Total spray coverage is essential for best results.
Yarrow	2.5-5	-	-	-	Apply to plant with 5-10cm of leaf. For best results cultivate 10 days after treatment. Use the low rate when a crop to be drilled is competitive to yarrow and higher rate when it is a poor competitor.

Add the Slikka™ wetter and penetrant at the suppliers recommended rates.
Slikka™ is a trade name of Etec CROP SOLUTIONS.

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CONDITIONS OF SUPPLY - Read before buying and using.

If these conditions are unacceptable, return at once to the retailer unopened within seven days and the purchase price will be refunded in full.

This product conforms to the chemical description on the label. However, to the full extent permitted by law, all warranties of any kind, expressed or implied, with respect to the fitness for any purpose (including crop yield/performance/injury, product effectiveness or any other consequence), condition, merchantable quality or any other matter, are expressly excluded. Buyers and users accept that PGG Wrightson Ltd and its distributors are not liable in any way (including in negligence, tort and equity) to you or anyone else for any loss, costs, loss of income, damage or injury whether direct, indirect or consequential, howsoever arising in connection with this product including without limitation storage, application, handling or use. PGG Wrightson Ltd's liability if any is limited solely to an amount equal to twice the purchase price of the product. To read PGG Wrightson Ltd's customer terms of trade that apply to purchases from PGG Wrightson Ltd, visit www.pggwrightson.co.nz"

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Batch:

Date of manufacture: